

A methodology to optimize site-specific field capacity and irrigation thresholds

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ABSTRACT

The determination of field capacity (FC), irrigation thresholds, and irrigation amounts is characterized by site-specific soil hydraulic properties (SHPs). This study, conducted in two zones (zone 1 and zone 2) delineated based on soil, topography, and historical crop yield in Alabama (USA), focused on determining zone-specific FC using negligible drainage flux (q_{fc}) criterion. The HYDRUS-1D model was used to optimize zone-specific SHPs using measured soil matric potential (h). The zone-specific FCs were determined using optimized and raw SHPs at 0.01 cm/day as q_{fc} . The results showed that the optimized FC at q_{fc} was at -39 kPa in zone 1 and raw FC was at -15 kPa. However, in zone 2, optimized FC was at -25 kPa and raw FC was at -59 kPa. To validate that optimized values are more accurate than raw values, a relationship between accumulated crop evapotranspiration (ET_c) and required irrigation amount was determined using optimized parameters (SHPs and FC) and showed a stronger correlation in both zones than using raw parameters (SHPs and FC). At flux-based FC, the optimized irrigation thresholds and amounts in zone 1 were -88 kPa and 20 mm, and raw irrigation threshold and amount were -58 kPa and 33 mm, respectively. In zone 2, the optimized irrigation thresholds and amounts were -45 kPa and 18 mm, and raw irrigation threshold and amount were -116 kPa and 14 mm, respectively. Therefore, using raw and benchmark FC can result in inefficient irrigation strategies. The proposed novel method of optimizing zone-specific FC and irrigation thresholds can help with adopting timely best irrigation management schemes in respective zones.

1. Introduction

Field Capacity (FC) criterion is important in defining the initial hydrologic state of soils, which affects both agronomic and hydrologic applications, and determining the effective soil water available to plants. Despite its significance, the definition of FC remains obscure for

well-defined hydrologic states of soils. Traditionally, a static soil matric potential (h) of -10 kPa for coarse-textured soils and -33 kPa for fine-textured soils has been used to represent FC (Rivers and Shipp, 1972). However, studies have indicated discrepancies and uncertainty associated with the static nature of FC (Liang et al., 2016), and these values change based on soil physical and hydraulic characteristics and soil

Abbreviations: FC, Field capacity; HCC, Hydraulic conductivity curve; SHP, Soil hydraulic property; SWD, Soil water depletion; SWRC, Soil water retention curve; q_{fc} , Negligible drainage flux; θ , Soil moisture/soil water content; h , Soil matric potential; h_{SWD} , Soil matric potential at SWD; h_{FC} , Soil matric potential at FC.

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profile depth considered, which suggests FC is dynamic (Assouline and Or, 2014; Lena et al., 2022; Nemes et al., 2011; Twarakavi et al., 2009). As the upper limit of soil water holding capacity changes dynamically, the available soil water will change accordingly, ultimately varying irrigation recommendations (Lena et al., 2022). Defining such limits is imperative to improving irrigation water-use efficiency and agricultural water management practices (de Jong van Lier, 2017; Irmak et al., 2016; Pinheiro et al., 2019; Lena et al., 2022; Romano et al., 2011) since it allows us to define irrigation timing and depth considering a soil water depletion (SWD). Therefore, an accurate determination of h at FC that accounts for site-specific soil characteristics is the first step when the objective is to implement irrigation best management practices and mitigate the negative impact associated with over- or under-application of irrigation water.

Due to high uncertainty, onerous, and time-consuming measurements of FC in the field and laboratory, the idea of internal drainage flux has been a topic of interest (Lena et al., 2022; Meyer and Gee, 1999; Twarakavi et al., 2009). Computer codes have been used to determine FC when the drainage flux at the bottom of a pre-defined soil profile reaches a negligible value. Schroeder et al. (1994) used HELP software to estimate FC at negligible drainage flux (q_{fc}) ranging from 0.001 cm/day to 0.1 cm/day depending on the application. Hillel (1998) recommended a value of 0.05 cm/day as q_{fc} to estimate FC. Meyer and Gee (1999) found consistency with the original range from 0.001 cm/day to 0.1 cm/day to interpret FC. Twarakavi et al. (2009) tested three q_{fc} values (0.001, 0.01, and 0.1 cm/day) using the HYDRUS-1D model to determine FC for homogeneous soil profile, and a value of 0.01 cm/day was recommended to represent FC across a range of USA soils. Although many studies have proposed the use of different q_{fc} values to represent FC, the impact of different q_{fc} values on irrigation calculations is hardly ever investigated. One of the few examples is the study conducted by Lena et al. (2022) in which different q_{fc} were tested to estimate the soil matric potential at FC for irrigation recommendations in Alabama, and the FC using 0.01 cm/day and 0.025 cm/day resulted best for irrigation recommendations vs accumulated crop evapotranspiration (ET_c) relationships in fine-textured and coarse-textured soils, respectively. This study showed the importance of site-specific evaluation of q_{fc} to reduce the uncertainties in FC determination and irrigation recommendations; however, the study did not optimize the FC and irrigation thresholds. Irrigation threshold refers to soil matric potential (h) at which irrigation is initiated. Further, Kumar et al. (2022a) optimized the soil hydraulic properties (SHPs) but did not optimize the FC to determine irrigation thresholds for irrigation scheduling and mentioned it as a limitation of the study. Therefore, we propose a method to optimize the FC and address the innovative solution for the research gap in the study investigated by Kumar et al. (2022a) and Lena et al. (2022).

The use of q_{fc} to estimate FC has been well accepted by the scientific community; however, the accurate determination of FC highly depends on well-defined and effective SHPs, i.e., soil water retention curve (SWRC) and hydraulic conductivity curve (HCC). Field or laboratory measurements of SHPs have been considered the standard methods since they provide site-specific information about soil physical characteristics, but these methods are time consuming, onerous, and include disparities in the measurements (Diamantopoulos et al., 2015). Additionally, a recent study indicated discrepancies between field observations and FC determined in the laboratory-based SHPs, which impacted the irrigation scheduling (Evelt et al., 2019). An alternative is to use estimated SHPs based on soil texture or pedo-transfer functions (Schaap et al., 2001), with the drawback of not being as precise as measured SHPs. One way to overcome the limitations regarding measured and estimated SHPs is the optimization of SHPs with h or soil water content (θ) measured using soil sensors in the field. Accurate determinations of SWRC and hydraulic conductivity (K) help in characterizing the soil water dynamics (Diamantopoulos et al., 2013; Iden et al., 2021), which is important in

irrigation scheduling. Therefore, optimizing the K - θ - h relationship is desirable in q_{fc} based determination of FC and irrigation scheduling to reduce uncertainty. In our knowledge, there has been a lack of research into the effects of optimized SHPs on determining FC and irrigation recommendations in actual field conditions. The accurate SHPs enable researchers and farmers to increase water use efficiency, improve crop yield, and maintain water resources by using improved irrigation management strategies (Jabro and Stevens, 2022). We hypothesize that optimized SHPs that also take into consideration the differences in site-specific properties in the crop field can not only improve the determination of FC but also the irrigation threshold values over the measured or estimated SHPs during the growing season, which can help with irrigation scheduling decisions. Thus, for site-specific management, it is best to delineate a crop field into different management zones based on soil texture, topography, historical crop yield, etc. (Kumar et al., 2022a) and determine zone-specific FC to avoid discrepancies by assuming one static FC throughout the field for irrigation water management.

Real-time h or θ monitoring, using sensors, in the soil profiles to schedule irrigation has been gaining attention during the last decade (Fontanet et al., 2020). Although this sensor-based irrigation can improve irrigation efficiency and reduce water stress (Irmak et al., 2010; Leininger et al., 2019), these improvements are only achieved if the K - θ - h relationship is accurately and precisely defined. Due to variability in soil hydraulic and physical characteristics, different irrigation thresholds have been adopted to trigger irrigation among the soil types. Leininger et al. (2019) recommended an h of -50 kPa for very-fine sandy loam textured soil in peanut field, and Irmak et al. (2016) recommended values ranging from -20 kPa to -25 kPa for fine sandy soils and from -90 kPa to -110 kPa for silt loam and silty clay loam soils in corn field. However, a single irrigation threshold has been adopted for the entire field and a single threshold does not account for site-specific variability in h or θ due to soil physical properties, topography, and soil water dynamics monitored within the field, and these values were recommended considering non-optimized SHPs. The determination of FC using q_{fc} has shown promise in determining irrigation thresholds. However, investigations to quantify the impact of zone-specific optimized SHPs and FC at q_{fc} on determination of zone-specific irrigation thresholds considering discrepancies in soil water dynamics monitored in the field are hardly ever conducted. Recently, Kumar et al. (2022a) optimized the irrigation thresholds using the soil textured based FC without considering zone-specific characteristics and indicated the limitation of not optimizing the FC. Similarly, studies (Irmak et al., 2016; Lena et al., 2022) used the SHPs determined in the laboratory or based on the soil texture assuming FC based on soil texture. These are "raw or observed SHPs" and these SHPs do not account for soil water dynamics in determining the irrigation thresholds and FC. Optimizing the raw or observed SHPs optimize the FC and result in optimized irrigation thresholds to schedule site-specific irrigation. Zone-specific differences in h or θ can be due to soil physical properties, topographic gradient, and vegetation (Kumar et al., 2022a). Hence, improved and optimized zone-specific SHPs using measured h or θ can help in determining upper limit of soil water holding capacity, consequently improving irrigation strategy during the growing season.

The specific objectives of this study were (1) to develop a modeling approach of optimizing zone-specific FC considering a q_{fc} criterion using the HYDRUS-1D model and investigate the impact of zone-specific optimized SHPs on the irrigation recommendation and (2) to investigate the effect of zone-specific optimized FC and SHPs on the zone-specific irrigation thresholds and irrigation amounts at soil water depletion (SWD). The study was conducted in two irrigated management zones of a delineated crop field. We used optimized and raw zone-specific SHPs in this investigation. To validate the FC simulations using optimized and raw SHPs, we estimated the relationship between accumulated crop evapotranspiration (ET_c) and required irrigation amounts

for irrigation recommendations using optimized and raw SHPs. In this article, we use term “raw SHPs” for the measurements obtained in the laboratory using HYPROP measurement systems (Kumar et al., 2022a) and “optimized SHPs” resulted after optimization using inverse HYDRUS-1D model. The analysis was conducted in two phases (1) inverse model development for optimization using input parameters (Sections 2.3 and 2.4 of materials and methods), and (2) negligible drainage flux model development for field capacity simulations and evaluation for irrigation recommendations (Sections 2.5.1 and 2.5.2 of materials and methods).

2. Materials and methods

2.1. Field site description and instrumentation

The study was performed in a 120 ha field ($34^{\circ} 43' 06.46''$ N and $87^{\circ} 23' 13.53''$ W) located in the Town Creek Watershed of the Tennessee Valley Region of North Alabama (Fig. 1). The detailed information about study site, irrigation management zones, irrigation system, irrigation scheduling (Lena et al., 2021a, 2021b), precipitation, and crops in the region are given in Kumar et al. (2022b). A brief description is given in this paper. The area is classified as subtropical humid (Cfa, Köppen climate classification) (Kottek et al., 2006) and has an annual average rainfall of about 1400 mm. The entire field was covered by a full circle of remotely controlled 625 m long Reinke center pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, NE, USA) with mid-elevation spray application (MESA) sprinklers fitted with pressure regulators.

Climate change is affecting the rainfall over the years (Takhellam-bam et al., 2023, 2022), which would alter time, erosion, and intensity of rainfall events. The distribution of rainfall over the growing season is important for irrigation management and for crop water demand (Kumar et al., 2021a, 2021b). Although crops might receive enough

rainfall by the end of the growing season, the time of rainfall is critical to reducing the crop water stress. Therefore, it is essential to plan irrigation according to the site-specific conditions and crop water demand. The field was delineated in three irrigation management zones (zone 1, zone 2, and zone 3) based on soil texture, topography, and historical yield data (Kumar et al., 2022a, 2022b) using Management Zone Analyst software (MZA 1.01, USDA-ARS, Columbia, MO, USA) (Fridgen et al., 2004). The slope in the field varied from 0% to 10%. The corn grain yield was monitored using a yield monitor mounted on the combine. We also generated a topographic wetness index (TWI) in the field (Beven, Kirkby, 1979), and TWI varied from 5.11 to 18. Two zones (zone 1 and zone 2 in Fig. 1), which were different in soil texture, topographic gradients (TWI and slope), and crop yield were selected for this study similar to Kumar et al. (2022a) (Fig. 1). Zone 1 had a lower slope, greater TWI, and greater crop yield as compared to zone 2, which had a greater slope, lower TWI, and lower crop yield. Zone 3 had properties between zone 1 and zone 2; therefore, it was not used in the study. Table 1 shows the average particle size distribution of zone 1 and zone 2. Based on particle size distribution of each zone, the soil profiles in each zone were relatively uniform. An array of three Watermark sensors (Irrometer Co., Riverside, CA, USA) were installed at multiple locations in both zones at 15 cm, 30 cm, and 60 cm soil depths during the growing season to monitor zone-specific h in the soil profile. The slurry was added to fill the holes and to ensure good contact between the sensors and the ground. The antennas and electronic boards were configured, and a base station was set up close to the field. In the 2018 growing season, sensors were installed on 16 May 2018 and removed on 12 August 2018. During the 2019 growing season, sensors were installed on 01 May 2019 and removed on 11 August 2019. The corn was planted on 28 March 2019 and harvested on 30 August 2019. During the 2018 growing season, corn was planted on 10 April and harvested on 03 September. An automated weather station (Vantage Pro2 Plus, Davis Instruments, Hayward, CA,

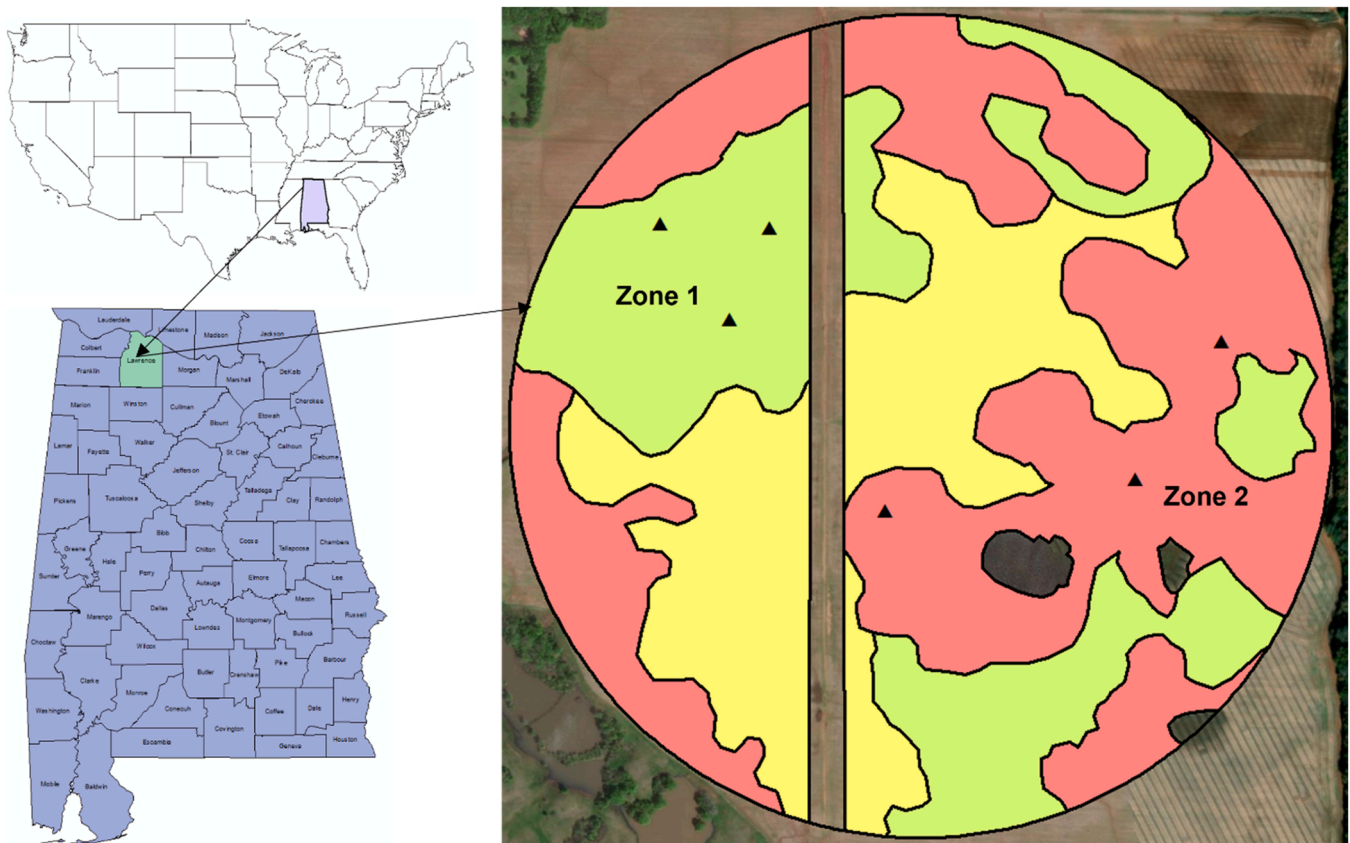


Fig. 1. Schematic diagram of the study field with management zones and sensor locations.

Table 1

Zone-specific soil properties in the crop field. (* indicate data retrieved from SSURGO 2022).

Zones ^a	Depth (cm)	Sand (%)	Silt (%)	Clay (%)	Texture	Soil Series*	Soil Family*
Zone 1	0–15	17.5	42.3	40.2	Silty Clay	Abernathy-Emory	Fine-silty, siliceous, active, thermic Oxyaquic Hapludults
	15–30	17.6	50.1	32.3	Silty Clay Loam		
	30–60	12.9	51.4	35.7	Silty Clay Loam		
Zone 2	0–15	31.9	32.8	35.3	Clay Loam	Decatur	Fine, kaolinitic, thermic Rhodic Paleudults
	15–30	30.1	29.6	40.3	Clay		
	30–60	29	24.4	46.8	Clay		

^a The zone 1 and zone 2 are delineated irrigation management zones within the field.

USA) was installed near the field to record meteorological parameters.

2.2. HYDRUS-1D model description

The HYDRUS-1D is a numerically approximate model which has been widely used to simulate soil water and nutrient dynamics (Glaesner et al., 2018) in different soil types and field conditions. The model uses an axisymmetric finite element code to solve Richard's equation numerically (Šimůnek et al., 2016). The HYDRUS-1D model has been used to simulate the soil water dynamics in multiple types of variably saturated soils in different crops such as corn, wheat, and paddy (Kumar et al., 2022a; Iqbal et al., 2020; Zhou and Zhao, 2019). The one-dimensional equation for water flow in an unsaturated flow medium is described by Eq. (1).

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S \quad (1)$$

where θ is soil water content [L^3L^{-3}]; h is soil matric potential [L]; t is time [T]; z is the vertical spatial coordinate [L]; K is unsaturated hydraulic conductivity [LT^{-1}]; and S is sink term in the water flow equation accounting for water uptake by roots [$L^3L^{-3}T^{-1}$].

The zone-specific SHPs were modeled using van Genuchten (1980):

$$\theta(h) = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{[1 + (\alpha|h|)^n]^m}; & h < 0 \\ \theta_s; & h \geq 0 \end{cases} \quad (2)$$

$$K(h) = \begin{cases} K_s S_e^l \left[1 - \left(1 - S_e^{\frac{1}{m}} \right)^m \right]^2; & h < 0 \\ K_s; & h \geq 0 \end{cases} \quad (3)$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (4)$$

θ_r is residual soil water content [L^3L^{-3}]; θ_s is saturated soil water content [L^3L^{-3}]; α [L^{-1}], m [-], and n [-] are shape parameters, where $m = 1 - 1/n$; l [-] is shape parameter in hydraulic conductivity function; K_s saturated hydraulic conductivity [LT^{-1}]; S_e is relative saturation or effective water content [-]. Eq. (2) is used for the conversion of h at any hydrologic state of soils into θ with the help of SWRC. Similarly, θ can be converted into h , as mentioned in the following 2.5.3 section of the methodology (Eq. (15)). The h and θ are interchangeable using SWRC for any hydrologic state of soils. Based on measurements h or θ , the user can convert one hydrologic component into another to fulfill the needs.

2.3. HYDRUS-1D model input parameters for inverse modeling

2.3.1. Determination of zone-specific raw soil hydraulic properties

Undisturbed soil cores were collected for SHPs in both zones using a stainless steel sampler ring (Meter Group, Pullman, WA, USA) of 5 cm in height and 8 cm diameter (251.33 cm³). To collect the soil samples, we used four tools including, soil sampling ring, sample ring insertion tool (Meter Group, Pullman, WA, USA), rubber mallet, and trowel. The

sampling ring comes with two plastic caps to cover the blunt and cutting edges of the ring. One side of the sampling has a blunt edge, and the other has a cutting edge. The blunt edge of the sampling ring was attached to the insertion tool, and a sharp edge was placed facing the soil surface, followed by hammering using the rubber mallet until the sampling ring fully penetrated the ground. Then, top surface of the sampling ring was covered with the sampling ring cap. The sampler rings were carefully removed by digging out surrounding soils using trowel to maintain the original soil properties, then the cutting edge was also covered with a sampling ring cap and transported to measure the SHPs (SWRC and HCC) in the laboratory (raw or observed SHP). After samples were brought to the lab, the cap was removed from the blunt edge, and covered with a cheesecloth. On top of the soil cores, a perforated tray was then positioned. Then, the soil cores were flipped, and cap was removed from the cutting edge too. The soil cores were placed in an empty container and the container was filled with distilled water to soak for 24 h or until the cores were fully saturated. While filling the container with water, it is necessary not to pour water on top of soil core to avoid the air trap. The water level in the container was kept up to the cutting edge of the ring to allow saturation from bottom to top. The cap was placed on the top of cutting edge to avoid the evaporation. A detailed procedure for sample preparation and experiment run is given in Shokrana and Ghane (2020). The ring was inserted into the hydraulic property analyzer (HYPROP 2, Meter Group, Pullman, WA, USA) for the determination of the wet end of the SWRC using the evaporation method (Peters et al., 2015). HYPROP measurements are limited by the cavitation point of the two tensiometers, and for this reason, water retention curves can be estimated from saturation until around pF 3. The pF is a logarithmic function of absolute h in cm.

$$pF = \log_{10}|h| \quad (5)$$

The h - θ data and weight of soil samples were automatically recorded by HYPROP-VIEW software (version 4.2.2.0, Meter Group, Pullman, WA, USA) every 30 min. After h - θ data measurements, the soil samples were removed from the hydraulic property analyzer and placed into an oven at 105 °C to determine the dry weight. After the determination of the wet end of the SWRC, the soil was removed from the sampler ring and four replicates of small disturbed samples were taken for the measurement of the dry end of the SWRC using a dew point hygrometer (WP4C PotentialMeter, Meter Group, Pullman, WA, USA). The soil sample was placed in the chamber of the dew point hygrometer to measure h for the dry end. The permanent wilting point (PWP) was measured at −1500 kPa in both zones. Soil samples were weighed immediately for wet weight and dried in the oven for determining dry weight. Detailed procedures for determining the wet and dry end of the SWRC are shown in Lena et al. (2022). The h - θ pairs measured by the hydraulic property analyzer and dew point hygrometer were input into HYPROP-FIT software (version 4.2.2.0, Meter Group, Pullman, WA, USA) to generate the SHPs. A single porosity model (Mualem, 1976; van Genuchten, 1980) characterized the SHPs to satisfy the relation in both zones. The generated SHPs in the laboratory are referred to as raw SHPs in the following sections.

2.3.2. Dynamic field measurements

The collected meteorological observations from installed weather station, including water flux (rainfall and irrigation), minimum and maximum temperatures, relative humidity, solar radiations, and wind speed, were input into the HYDRUS-1D model. We recorded the growth stages and plant heights in both zones to determine the associated crop coefficients (K_c) using Irmak (2017) to determine crop evapotranspiration (ET_c). The leaf area index (LAI) using LAI-2200 Plant Canopy Analyzer (LI-COR Environmental, Lincoln, NE, USA) on the 57th, 72nd, 92nd, and 128th day after planting at four locations in each zone with five replications of each location was also measured (Kumar et al., 2022a). These crop data were also provided to the HYDRUS-1D model for each zone. To input collected meteorological and crop data, we selected time-variable boundary conditions in the HYDRUS-1D model.

2.3.3. Initial and boundary conditions

The user can define the initial conditions based on h or θ in the HYDRUS-1D model. We defined initial conditions as h in both the zones for our study since we measured h in both the zones. Since the study was a field study and the surface layer was subjected to the atmosphere, we selected the upper boundary condition as atmospheric boundary conditions with surface runoff in the HYDRUS-1D model. From the meteorological data, HYDRUS-1D calculates reference evapotranspiration (ET_o) using Penman-Montheith equation. HYDRUS-1D does not consider ET_o to be a single process. HYDRUS-1D partitioned ET_o into potential evaporation and potential transpiration using Beer's law (Ritchie, 1972) and LAI:

$$T_p = ET_o(1 - e^{-k^*LAI}) \quad (6)$$

$$E_p = ET_o(e^{-k^*LAI}) \quad (7)$$

Where T_p is potential transpiration [LT^{-1}], E_p is potential evaporation [LT^{-1}], LAI is leaf area index, and k is radiation extinction [-] by canopy.

The lower boundary condition was selected as a free drainage boundary condition at 100 cm soil depth, indicating no interaction with the groundwater. Using boundary conditions at 60 cm in HYDRUS-1D for simulations, it introduces boundary condition effects, meaning that water crossing this level would never be available for uptake. In reality, there may be some upward flow from the 60–100 cm soil layer to the root zone if the soil is drying out. Therefore, free drainage boundary condition is a good approximation if the water table is relatively deep, and it does not affect water dynamic in the 100 cm profile. Also, the use of a 200 cm profile required more computation time and resources in numerical modeling and did not result in different outcomes than using a 100 cm profile. This was valid for the simulated zones in this study. Depending on crop and site information, the root zone depth and lower boundary conditions can vary, and the user must be careful in defining root zone depth and lower boundary conditions since they can affect the root water uptake. Such model set up has widely been used in previous studies conducted in field conditions (Kumar et al., 2022a; Lena et al., 2022; Mokari et al., 2019). The observation nodes were defined at the same depths of 15 cm, 30 cm, and 60 cm, where Watermark sensors were installed in both zones.

2.3.4. Root water uptake

HYDRUS-1D model provides the flexibility for users to select the root water uptake module depending on the crop that allows to calculate the actual transpiration relatively to the potential transpiration. The HYDRUS-1D has inbuilt crop modules for different crops, including corn, grass, tomato, etc. The sink term, S , in Richard's equation was computed using an equation given by Feddes et al. (1978).

$$S = \alpha(h)\beta(z)T_p \quad (8)$$

where $\alpha(h)$ [-] is a dimensionless response function of h ($0 < \alpha(h) < 1$) which accounts for root water uptake stress; $\beta(z)$ is normalized water

uptake distribution [L^{-1}]; T_p is a potential transpiration rate [LT^{-1}].

Water stress causes root water uptake reduction, $\alpha(h)$, can be explained as:

$$\alpha(h) = \begin{cases} 0; h > h_1 \text{ or } h < h_4 \\ \frac{h - h_1}{h_2 - h_1}; h_1 > h > h_2 \\ 1; h_2 \leq h \leq h_3 \\ \frac{h - h_4}{h_3 - h_4}; h_3 > h > h_4 \end{cases} \quad (9)$$

where h_1 is called anaerobiosis point, i.e., water uptake considered to be zero close to the saturation ($h > h_1$); h_4 is pressure head at the wilting point and water uptake is also considered to be zero when $h < h_4$. Maximum root water uptake happens between h_2 and h_3 , and it decreases or increases linearly between the pressure heads h_3 and h_4 or h_1 and h_2 , respectively (Fig. 2). Since 95% of the corn roots are linearly distributed within the 0–60 cm soil profile, and for this reason, model assumed linear root distribution function to a depth of 60 cm. Depending on the crop information, HYDRUS-1D model then assumes a linear interpolation with time. The parameters in Wesseling (1991) were selected for the root water uptake of corn in both zones.

2.4. Inverse modeling for SHP optimization and goodness of fit analysis

We used inverse modeling to optimize the SHPs using measured h from the array of three Watermark sensors at 15 cm, 30 cm, and 60 cm soil depths to simulate the soil water dynamics in each zone. After model setup with input parameters, HYDRUS-1D model was coupled with the Shuffled Complex Evolution (SCE-UA) algorithm for global optimization of SHPs (Diamantopoulos et al., 2012; Duan et al., 1994; Qu et al., 2014) in the zone 1 and zone 2. Due to fairly uniform soil profiles and textural homogeneity within the soil depths of each zone as mentioned by Kumar et al. (2022a), we used a single-layer zone-specific raw SHPs in each zone to build HYDRUS-1D model and optimized with multi-depth measured h . An objective function (Obj) was defined for minimization in the SCE-UA algorithm for each observation at each depth.

$$Obj = \min \sum_{j=1}^{nm} [pF_j - \widehat{pF}_j(x)]^2 \quad (10)$$

where $pF_j = [pF_{15cm}, pF_{30cm}, pF_{60cm}]$ is a vector of daily observations of h as pF values; $\widehat{pF}_j = [\widehat{pF}_{15cm}, \widehat{pF}_{30cm}, \widehat{pF}_{60cm}]$ is a vector contains daily HYDRUS-1D predictions of h as pF values; $x = (\theta_r, \theta_s, \alpha, n, K_s, l)$ is a vector of soil hydraulic parameters used in the van Genuchten Mualem model, θ_r is residual soil water content [L^3L^{-3}], θ_s is saturated soil water content [L^3L^{-3}], α [L^{-1}], m [-], and n [-] are shape parameters, l [-] is shape parameter in hydraulic conductivity function, K_s is the saturated hydraulic conductivity [LT^{-1}], j is the measurement time, and nm is the total number of measurements at each soil depth. To fully use the zone-specific SHPs, we consider all parameters of the van Genuchten Mualem model to optimize and simulate soil water dynamics accurately in both zones. The model was set up from 05 April 2018–30 August 2019 until harvesting the crop. From 05 April 2018 until 30 April 2019, the model was trained to set initial conditions to run the simulations for the optimization of SHPs.

The performance of the HYDRUS-1D model was checked using different statistical indices of goodness of fit. Root mean square error (RMSE), index of agreement (k), and relative root mean square error (RRMSE) were used in this study.

$$RMSE = \sqrt{\frac{1}{nm} \sum_{j=1}^{nm} (\widehat{pF}_j - pF_j)^2} \quad (11)$$

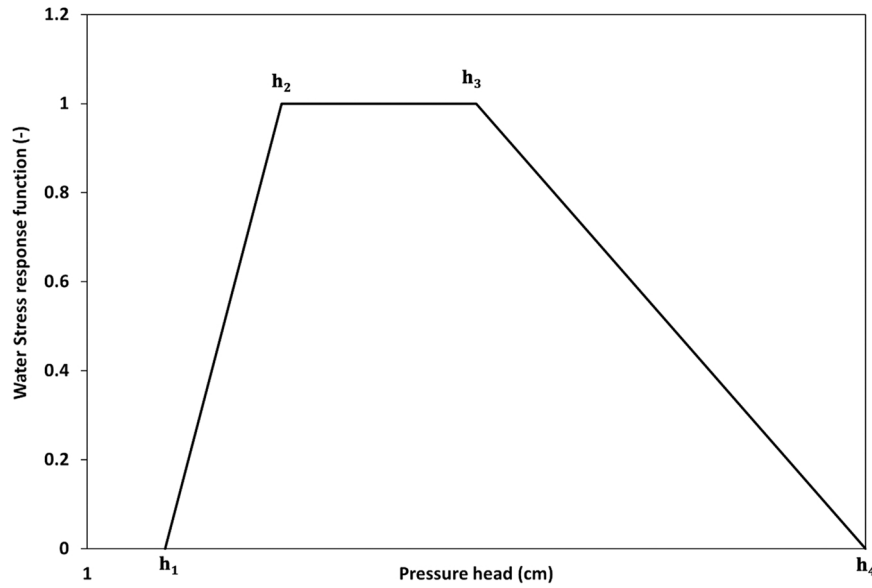


Fig. 2. Schematic diagram of root water uptake stress response function. h_1 is anaerobiosis point, h_4 is the wilting point pressure head, and h_2 and h_3 are pressure heads where water uptake was considered optimal.

$$d = 1 - \frac{\sum_{j=1}^{nn} (\widehat{pF_j} - pF_j)^2}{\sum_{j=1}^{nn} (|\widehat{pF_j} - pF_{avg}| + |pF_j - pF_{avg}|)^2} \quad (12)$$

$$RRMSE = \frac{\sqrt{\frac{1}{nn} \sum_{j=1}^{nn} (\widehat{pF_j} - pF_j)^2}}{pF_{avg}} \quad (13)$$

where nn is the total number of observations, $\widehat{pF_j}$ and pF_j are simulated and observed pF values, respectively, and pF_{avg} is the average observed pF value. The lower the $RMSE$ value, the better the performance of the model to predict the data in terms of its absolute error. The d value ranges from 0 to 1, and a value close to 1 indicates a better performance of the model. The $RRMSE$ indicates the performance of the model is excellent when $RRMSE < 10\%$, good if $10\% \leq RRMSE \leq 20\%$, fair if $20\% < RRMSE \leq 30\%$, and poor if $RRMSE > 30\%$.

2.5. Negligible drainage flux-based field capacity simulations and evaluation of flux-based field capacity for irrigation recommendations

2.5.1. Model development and simulations for negligible drainage flux-based field capacity

Once the optimized SHPs (from inverse modeling in HYDRUS-1D in Section 2.4) and raw SHPs (from HYPROP 2 in Section 2.3.1) were generated in both zones, the simulations of FC using the q_{fc} criterion were performed using the HYDRUS-1D model (Šimůnek et al., 2016) in both zones. Now, we had two sets of SHPs, optimized and raw, in each zone. Since the soils in zone 1 and zone 2 represented fine textures category, we adopted a q_{fc} value of 0.01 cm/day to simulate the FC in both zones, following the recommendation of Lena et al. (2022) for fine-textured soils in Alabama. We setup another HYDRUS-1D model for estimating h at FC in zone 1 and zone 2. For the model setup of FC simulations, the upper boundary condition was fixed as constant zero flux, the lower boundary condition as free drainage, and the initial condition was defined as h of -0.98 kPa (10 cm of water). The model setup was kept same as recommended in previous studies for FC simulations (Lena et al., 2022; Twarakavi et al., 2009). The zone-specific optimized and raw SHPs were used as input parameters in the HYDRUS-1D model for the flow of flux to determine h at FC in the

100 cm soil profile. Therefore, two HYDRUS-1D models, one for optimized SHPs and the other for raw SHPs in each zone, were setup for zone-specific FC simulations. When the bottom of the soil profile reached the desired q_{fc} of 0.01 cm/day, the profile was assumed to be at the FC condition. The FC determined using optimized SHPs was named as optimized FC, and the FC using raw SHPs was named as raw FC. After FC simulations using optimized and raw SHPs in each zone, we evaluated the impact of these optimized and raw FC on irrigation recommendations in each zone. Additionally, we also studied the impact of FC on determination of irrigation thresholds.

2.5.2. Negligible drainage flux criteria evaluation for zone-specific irrigation recommendations using optimized and raw SHPs

To prove if optimized FC and SHPs are more accurate than raw FC and SHPs, we developed the relationship between accumulated ET_c and required irrigation amounts for h_{FC} at optimized and raw FC. The ET_c was calculated using crop coefficient (Allen et al., 1998) as:

$$ET_c = K_c * ET_0 \quad (14)$$

where K_c is crop coefficient and ET_0 is reference evapotranspiration, and K_c was determined using Irmak (2017). These calculations were performed outside the HYDRUS-1D modeling environment to generate irrigation recommendations. Based on FC recommendations using a negligible drainage flux of 0.01 cm/day for zone-specific irrigation management, we determined the irrigation recommendation for optimized and raw zone-specific SHPs in this study. To evaluate the impact of q_{fc} of 0.01 cm/day using zone-specific optimized and raw SHPs on irrigation recommendations, we implemented an analysis using observed h data in each zone. We used three depths of installed sensors where h measurements were recorded during the growing season. For this analysis, the periods with no rainfall events were selected to calculate the required irrigation water as the amount of water required to replenish h measured with sensors back to h_{FC} (soil matric potential at FC). The required irrigation water was calculated based on h_{FC} and θ_{FC} determined using the q_{fc} of 0.01 cm/day. We also calculated accumulated ET_c for the same periods in which irrigation amounts were calculated. The daily ET_c values were calculated using Eq. (14). To calculate accumulated ET_c for the validation of this method, we selected the periods when there was a crop (minimum bare soil evaporation) in the field similar to Lena et al. (2022).

The accumulated ET_c values were calculated as the sum of daily ET_c values for the selected interval between irrigations or rainfall. For comparisons, a regression model was fit between required irrigation amounts based on 0.01 cm/day as q_{fc} and accumulated ET_c for the selected intervals in both zones using optimized and raw SHPs to validate optimized parameters (SHPs and FC) are more accurate than raw parameters.

2.5.3. Determination of irrigation thresholds and amounts

The SWD-based irrigation method is the most commonly adopted technique to determine irrigation thresholds or amounts to schedule irrigation which can reduce crop water stress and increase the crop yield (Kisekka et al., 2016). The irrigation thresholds and amounts can be determined using a predefined value of SWD in the following equations. Depending on the soil water depletion in the crop fields, irrigation thresholds and irrigation amounts can be determined for irrigation scheduling during the growing season.

$$|h|_{SWD} = \frac{\left\{ \left[\frac{\theta_s - \theta_r}{\theta_{SWD} - \theta_r} \right]^{1/m} - 1 \right\}^{1/n}}{\alpha} \quad (15)$$

$$\theta_{SWD} = \theta_{FC} - [SWD(\theta_{FC} - \theta_{PWP})] \quad (16)$$

$$I_{SWD} = \sum (\theta_{FC} - \theta_{SWD})Z \quad (17)$$

where $|h|_{SWD}$ is irrigation threshold as soil matric potential (h) at pre-

defined SWD [L]; θ_{SWD} is the soil water content at predefined SWD [L^3L^{-3}]; θ_{FC} is the soil water content at FC [L^3L^{-3}]; θ_s is the saturated soil water content [L^3L^{-3}]; θ_r is the residual soil water content [L^3L^{-3}]; α [L^{-1}], m [-], and n [-] are shape parameters; θ_{FC} is the soil water content at FC [L^3L^{-3}]; θ_{PWP} is the soil water content at PWP [L^3L^{-3}]; I_{SWD} is the irrigation amount [L]; Z is the thickness of the soil layer [L]. The soil water content at FC using the q_{fc} criterion for each SHP and the soil water content at PWP (−1500 kPa) in each zone were used to determine the upper and lower limits of water holding capacity to derive irrigation thresholds and amounts. The upper limit was set as 0% SWD (at FC), while the lower limit was set as 100% (at PWP).

3. Results and discussion

3.1. Goodness of fit analysis of inverse HYDRUS-1D modeling

The raw SHPs from both zones showed differences in the pattern of SWRC and HCC (Fig. 3). The differences in the pattern of SWRC and HCC indicated the need for optimization using the observed data in the field conditions. From Fig. 3 towards the lower pF end of the raw SWRCs, the raw SWRC in zone 2 started at lower soil water content as compared to raw SWRC in zone 1, which can also be observed at pF= 0 ($h=0$), i.e., saturation. When pF increases, raw SWRCs in both zones intersected at around pF= 2 ($h=-9.8$ kPa) and acquired reversed pattern of the SWRCs towards higher pF compared to the pattern at lower pF in both zones (Fig. 3). At pF= 4.2 ($h=-1500$ kPa) using raw SWRCs, the zone 2 had a higher soil water content (20%) than the soil water content (14%)

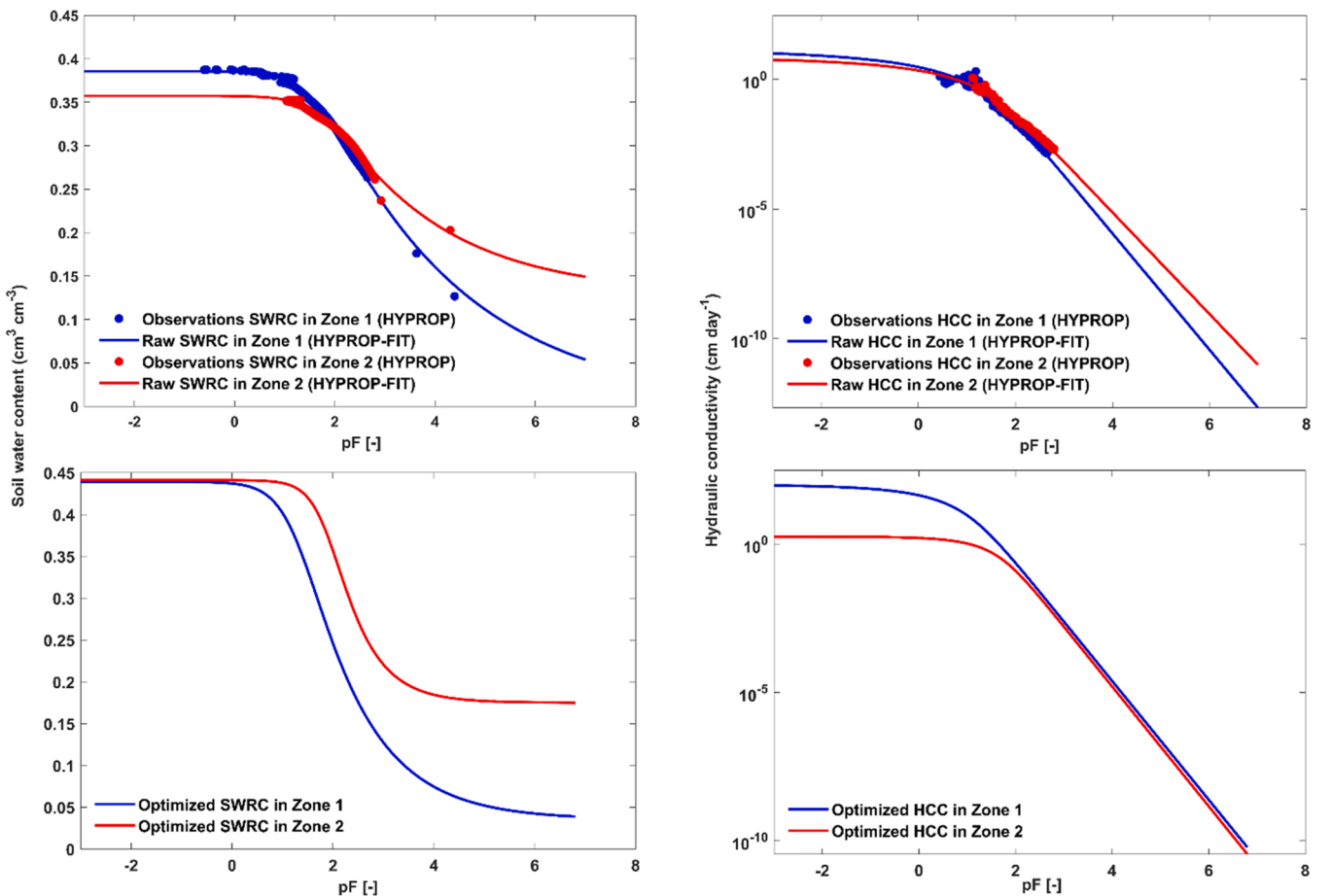


Fig. 3. Zone-specific raw (observed) SHPs; SWRC (top left) and HCC (top right) showing measurements with HYPROP measurement system and fitted with HYPROP-FIT software. HYPROP produced around 500 observations to generate SWRC and HCC. Zone-specific optimized SHPs; SWRC (bottom left) and HCC (bottom right) obtained using inverse modeling in HYDRUS-1D. The pF is logarithmic function of absolute soil matric potential (h). The zone 1 and zone 2 are the delineated irrigation management zones within the field. SHPs: soil hydraulic properties, SWRC: Soil water retention curves, HCC: hydraulic conductivity curve.

in zone 1. However, after inverse modeling using HYDRUS-1D model, optimized SWRCs in both zones had identical soil water content (44%) at saturation with a $pF = 0$ (Fig. 3), and optimized SWRCs in both zones never intersected. The optimized SWRCs had the same pattern from lower pF to higher pF ends with higher soil water content in zone 2 compared to zone 1 throughout the patterns of SWRCs (Fig. 3). For example, at $pF = 4.2$ with optimized SWRCs, the soil water content in zone 2 was 18%, and the soil water content in zone 1 was 7%, which showed the differences in zone-specific retention characteristics. Although raw SHPs showed the importance of zone-specific SHPs between the zones, the raw SHPs did not consider the actual soil water dynamics within the soil profiles in the field conditions to address the significance of the observed h in each zone and ignored the soil water movement within the soil profiles. Therefore, we need to consider spatial variability in the raw SHPs between the zones for better irrigation management and adopt zone-specific SHPs. Since both zones had distinct patterns of SHPs, which means invariable SHPs cannot be assumed throughout the study field for irrigation management. For example, if we assume that raw SHPs of zone 1 represents the entire field, the soil water content at $pF = 3$ in the field was 23% for the entire study field. Therefore, it reflects that if irrigation needs to be applied, it will be applied at one rate throughout the study field, which can result in over-irrigation in some parts and under-irrigation in others due to zone-specific differences in SHPs. However, if we consider zone-specific optimized SHPs, the soil water content at $pF = 2$ was 25% in zone 1 and 36% in zone 2. This reflects that zones will have different soil water content at the same soil matric potential, which can lead to different rates of irrigation. Hence, based on the soil water content differences at a particular pF reflects that the considering uniform SHPs within the field can be a flawed irrigation strategy due to zone-specific variability in the pattern of SHPs.

The performance of the HYDRUS-1D model was tested at multiple locations in each zone. The results showed that the simulated pF fit well with measured values of pF at all three soil depths in both zones (Table 2). The one-layer optimized HYDRUS-1D model is a simplified version using single layer SHPs of the complex system, which can be used to simulate soil water dynamics at all three different layers. In our study, we found d varied from 0.78 to 0.86 with an average of 0.81 in zone 1 and from 0.62 to 0.89 with an average of 0.78 in zone 2. The RRMSE was less than 0.2% in both zones at all the soil depths. We achieved RMSE ranging from 1.66 cm to 2.57 cm (0.22–0.41 as pF) with an average of 2.19 cm (0.34 as pF) in zone 1 and from 1.74 cm to 3.55 cm (0.24–0.55 as pF) with an average of 2.39 cm (0.37 as pF) in zone 2 in the entire soil profile (Table 2). On comparing the inversing modeling performance of HYDRUS-1D with previous multilayer modeling studies, Li et al. (2014) achieved the minimum average RMSE of 2.61 cm (0.42 as pF) using an individual SWRC for each soil layer from 0 to 100 cm at 20 cm depth increments. Wyatt et al. (2017) found RMSE ranged from 897 cm to 3388 cm (2.95–3.53 as pF) with a higher RMSE in a deep soil layer to estimate groundwater recharge at four various sites in Oklahoma. Mokari et al. (2019) achieved a d varying from 0.44 to 0.73 to simulate soil water movement. However, these

studies did not consider all van Genuchten parameters to optimize the SHPs. Based on the statistical indices to evaluate HYDRUS-1D model, the performance was evaluated as excellent to optimize zone-specific SHPs since our results were consistently better than previous studies. The variation in the model performances in both zones can be due to differences in the SHPs and soil water retention behavior of the soils as we found different soil water content at same pF in both zones. Additionally, since a high slope area increases the chances of sub-surface lateral water movement compared to low slope areas (Huat et al., 2006), the low slope in zone 1 and high slope in zone 2 can affect the soil water dynamics in both zones and one-dimensional models cannot measure lateral movement of soil water. Kumar et al. (2022a) found differences in root water uptake between two zones affected the crop characteristics as a greater plant height and LAI in zone 1 than in zone 2. The recorded plant heights and LAI values that were greater in zone 1 (maximum LAI of $4.13 \text{ cm}^2 \text{ cm}^{-2}$ with corresponding plant height of 225 cm) than in zone 2 (maximum LAI of $3.34 \text{ cm}^2 \text{ cm}^{-2}$ with corresponding plant height of 195 cm). The zone 1 had silt type texture in the soil profile; however, zone 2 had clay texture. The clay texture has maximum water holding capacity and minimum water releasing property (Çakir and Cangir, 2021). On the other hand, silt type soils have slightly less water holding capacity than clay but have greater water releasing property than clay soils. Similar outcomes of differences in water holding capacity were noticed in the soil of silt texture and clay texture in Turkey (Çakir, 2019; Çakir and Cangir, 2021).

3.2. Assessment of negligible drainage flux criteria for irrigation recommendations using optimized and raw SHPs

The relationship between required irrigation and accumulated ET_c was determined (Fig. 4a for zone 1 and Fig. 4b for zone 2) for zone-specific optimized and raw FC (Table 3) using zone-specific optimized and raw SHPs (Fig. 3). A linear model was fit to best represent the relationships for optimized and raw FC values in both zones. In zone 1, a coefficient of determination (R^2) of 0.88 was achieved between accumulated ET_c and required irrigation for optimized FC and SHPs; however, an R^2 of 0.50 for raw FC and SHPs. Similarly, an R^2 of 0.74 was achieved with optimized FC and SHPs in zone 2, and an R^2 of 0.33 for raw FC and SHPs. These relationships support the hypothesis of optimized FC and SHPs are more effective to use in irrigation recommendation than raw FC and SHPs in both zones since optimized ones consider soil water dynamics in determining the irrigation thresholds and amounts. The discrepancies in irrigation amount reflect the effect of determined FC and adoption of optimized and raw SHPs in each zone. With smaller values of h_{FC} , the amount required for irrigation amount will be lower to replenish the soil water content back to FC from a given h since h_{FC} gets closer to zero. Also, the higher values of h_{FC} will require a higher amount of irrigation water to replenish the soil water content back to FC from a given h since h_{FC} gets farther from zero. Based on the relationship between accumulated ET_c and required irrigation using optimized and raw FC and SHPs, we found that optimized FC and optimized SHPs resulted in a stronger correlation than raw FC and raw

Table 2

Goodness of fit statistics of inverse HYDRUS-1D model to simulate soil matric potential in both zones. The total 927 observations were used to compute the statistics.

Zones ^a	Performance statistics ^b	0–15 cm			15–30 cm			30–60 cm		
		RMSE (cm)	d	RRMSE (%)	RMSE (cm)	d	RRMSE (%)	RMSE (cm)	d	RRMSE (%)
Zone 1	Mean	2.29	0.82	0.1	1.95	0.84	0.1	2.40	0.78	0.1
	Min.	2.14	0.78	0.1	1.66	0.83	0.1	2.24	0.78	0.1
	Max.	2.40	0.85	0.2	2.14	0.86	0.1	2.57	0.79	0.2
Zone 2	Mean	2.14	0.83	0.1	2.40	0.79	0.2	2.51	0.71	0.1
	Min.	1.86	0.75	0.1	2.24	0.76	0.1	1.74	0.62	0
	Max.	2.57	0.89	0.2	2.51	0.81	0.2	3.55	0.84	0.2

^a The zone 1 and zone 2 are delineated irrigation management zones within the field

^b RMSE is root mean square error, d is index of agreement, and RRMSE is relative root mean square error.

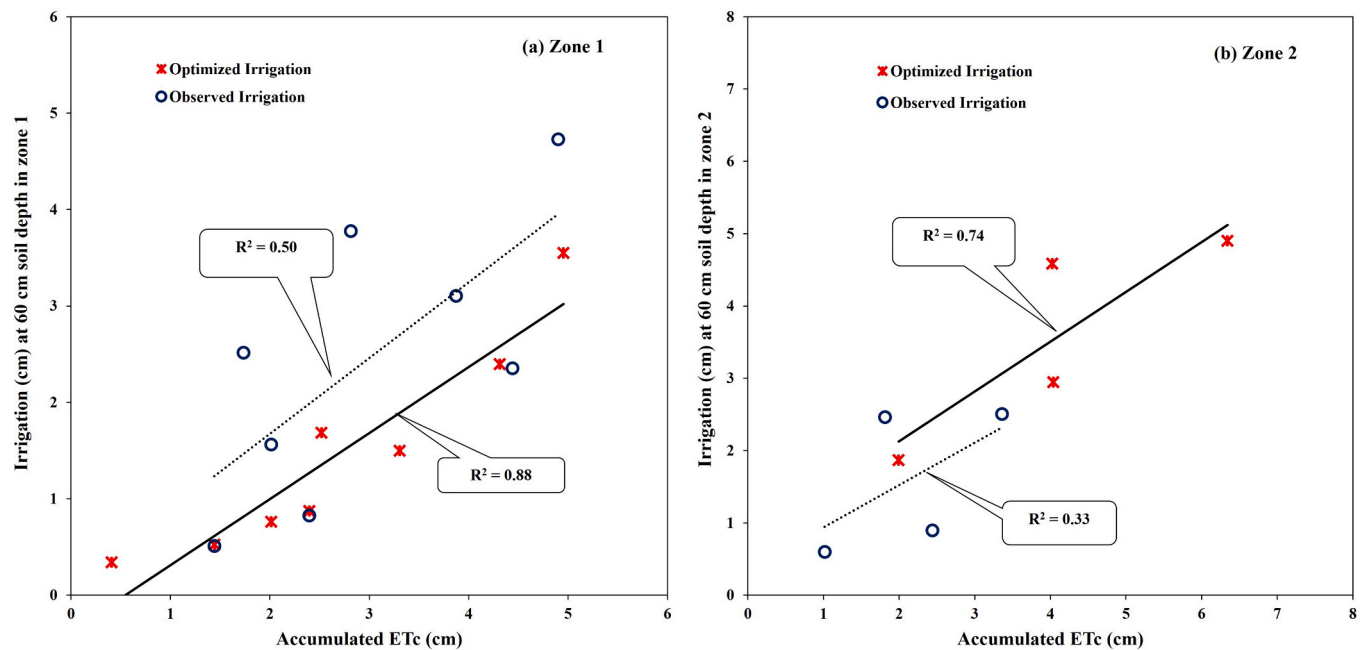


Fig. 4. Relationship between the accumulated crop evapotranspiration (ET_c) and required irrigation in a soil profile of 60 cm depth using 0.01 cm/day as q_{fc} for field capacity simulations using zone-specific optimized and raw SHPs in (a) zone 1 and (b) zone 2. R^2 is the coefficient of determination for the fitted data.

Table 3

Determined field capacity values using 0.01 cm/day as negligible drainage flux for soil hydraulic properties (optimized and Raw) and benchmark field capacity at 35% soil water depletion.

Zones ^a	Optimized FC (kPa)	Raw FC (kPa)	Benchmark FC (kPa)
Zone 1	-39	-15	-33
Zone 2	-25	-59	-33

^a The zone 1 and zone 2 are delineated irrigation management zones within the field.

SHPs in both zones (Fig. 4a for zone 1 and Fig. 4b for zone 2), which means that irrigation water is being used more efficiently.

The results found in this study suggest that the use of zone-specific optimized FC for irrigation recommendations can potentially improve the irrigation management practices. Therefore, we recommend optimizing zone-specific FC values (Table 3) using optimized SHPs (Fig. 3) for determining irrigation recommendations. The FC determination using raw SHPs or assuming benchmark FC may not be accurate in irrigation management due to discrepancies between monitored h in the fields and laboratory experiments to develop SHPs (Fig. 3). Such discrepancies between FC determined in the lab and field soil water measurements have been discussed by Evett et al. (2019). However, we do not suggest obliterating the determination of FC using such SHPs methods. The optimized threshold values (Table 4) for irrigation triggering showed the differences between the zones and resulted in promising results of zone-specific irrigation demands, which might

result in the variable time of irrigation threshold arrival and frequency of irrigations in the zones. Using this inverse modeling approach to optimizing FC, we can determine effective irrigation thresholds and amounts to trigger irrigation, which reduces the chance of flawed irrigation using raw SHPs and soil textured-based FC. Since FC is the upper limit for irrigation and critical in determining the irrigation thresholds for sensor-based irrigation, it is necessary to consider the soil water dynamics to optimize the SHPs and site-specific FC.

3.3. Flux-based field capacity estimation using HYDRUS-1D

The determined FC values using different SHPs are tabulated in Table 3. On comparison between optimized and raw FCs, the optimized FC was at -39 kPa and the raw FC was at -15 kPa in zone 1 (Table 3). However, in zone 2, the optimized FC was at -25 kPa and the raw FC was at -59 kPa. A similar trend of FC was also found by Lena et al. (2022) using multilayered raw SHPs for fine-textured soil types. However, Lena et al. (2022) did not optimize SHPs and FC values. These differences between optimized and raw FC (Table 3) in each zone resulted since raw FC did not account for soil water dynamics and optimized FC did account for soil water dynamics using h observations in each zone. The impact of optimized FC and raw FC in each zone were noticed on irrigation recommendation in Figs. 4a and 4b. Also, the differences in zone-specific optimized FC values show the differences in dynamic soil water movement processes that occurred in the actual field conditions, which were captured by zone-specific h measurements recorded during the growing season and incorporated in the inverse HYDRUS-1D modeling for optimization (Table 2). Measuring FC in the

Table 4

Irrigation thresholds (kPa) and irrigation amounts (mm) at 35% soil water depletion for negligible drainage flux-based field capacity and benchmark field capacity.

	Irrigation threshold (h_{35})				Irrigation amounts (mm)			
	Zone 1		Zone 2		Zone 1		Zone 2	
	Flux	Benchmark	Flux	Benchmark	Flux	Benchmark	Flux	Benchmark
Optimized	-88	-76	-45	-59	20	22	18	15
Raw	-58	-102	-116	-70	33	26	14	19

The zone 1 and zone 2 are delineated irrigation management zones within the field.

field condition is impossible to control the external factors and know the drainage flux in the sub-surface, which was achieved in the HYDRUS-1D. The irrigation recommendations reflected this impact of optimized FC in each zone (Figs. 4a and 4b). The differences in zone-specific FC values considerably affect the irrigation water management to meet the water demands required for a crop in each zone (Figs. 4a and 4b). Additionally, the optimized FC values compared to benchmark FC (-33 kPa) were different in both zones. The optimized FC was at -39 kPa in zone 1 and -25 kPa in zone 2 compared to the benchmark FC at -33 kPa, which was the same in both zones, shows that the benchmark FC did not account for zone-specific characteristics for soil water dynamics during the growing season. The differences in zone-specific FC can result in differences in the irrigation thresholds, and ultimately the time to trigger irrigation during the growing season.

Defining irrigation timing and amount is highly dependent on the upper limit of soil water holding capacity; therefore, different FC values will result in differences in irrigation recommendations. The differences in determined FC using optimized and raw SHPs suggested that irrigation recommendations can be improved when zone-specific optimized SHPs are used for optimizing FC with numerical modeling, ultimately promoting irrigation best management practices. The optimized FC in zone 1 and zone 2 have different values, which indicate the difference in the upper limits of soil available water. Assuming a static FC value of -33 kPa as the benchmark FC throughout the field for irrigation scheduling may not be accurate for efficient irrigation water management and can be flawed for irrigation recommendations (Nemes et al.,

2011). This assumption can over-or under-estimate the zone-specific irrigation thresholds, which can result in over-or under-irrigation in the zones (Table 4). Therefore, determining accurate zone-specific FC values can enable precise and zone-specific irrigation scheduling with improved irrigation thresholds in each zone.

3.4. Determination of zone-specific irrigation thresholds and amounts based on drainage flux FC criterion

The optimized irrigation thresholds and amounts were determined with variation in SWD to trigger irrigation in both zones (Fig. 5). In zone 1, we found that the optimized irrigation thresholds (absolute) were greater (Fig. 5a) and optimized irrigation amounts were lower (Fig. 5c) than raw irrigation thresholds and amounts, respectively, due to differences in SHPs and FC. The optimized irrigation amounts were always greater in zone 1 than zone 2 (Figs. 5c and 5d). The trend of the optimized and raw irrigation threshold curves indicated that the two curves intersect at SWD of greater than 70% in zone 1 (Fig. 5a), which show the reduced differences in the upper and lower limits with the shape of optimized and raw SHPs due to higher depletion. In zone 2, the optimized irrigation thresholds (absolute) were lower than the raw irrigation thresholds (Fig. 5b) and the optimized irrigation amounts were greater than the raw irrigation amounts (Fig. 5d). Comparing optimized irrigation thresholds of both zones, the optimized irrigation thresholds (absolute) were always greater in zone 1 than zone 2. Differences in the pattern of irrigation threshold and amount showed inaccuracy in

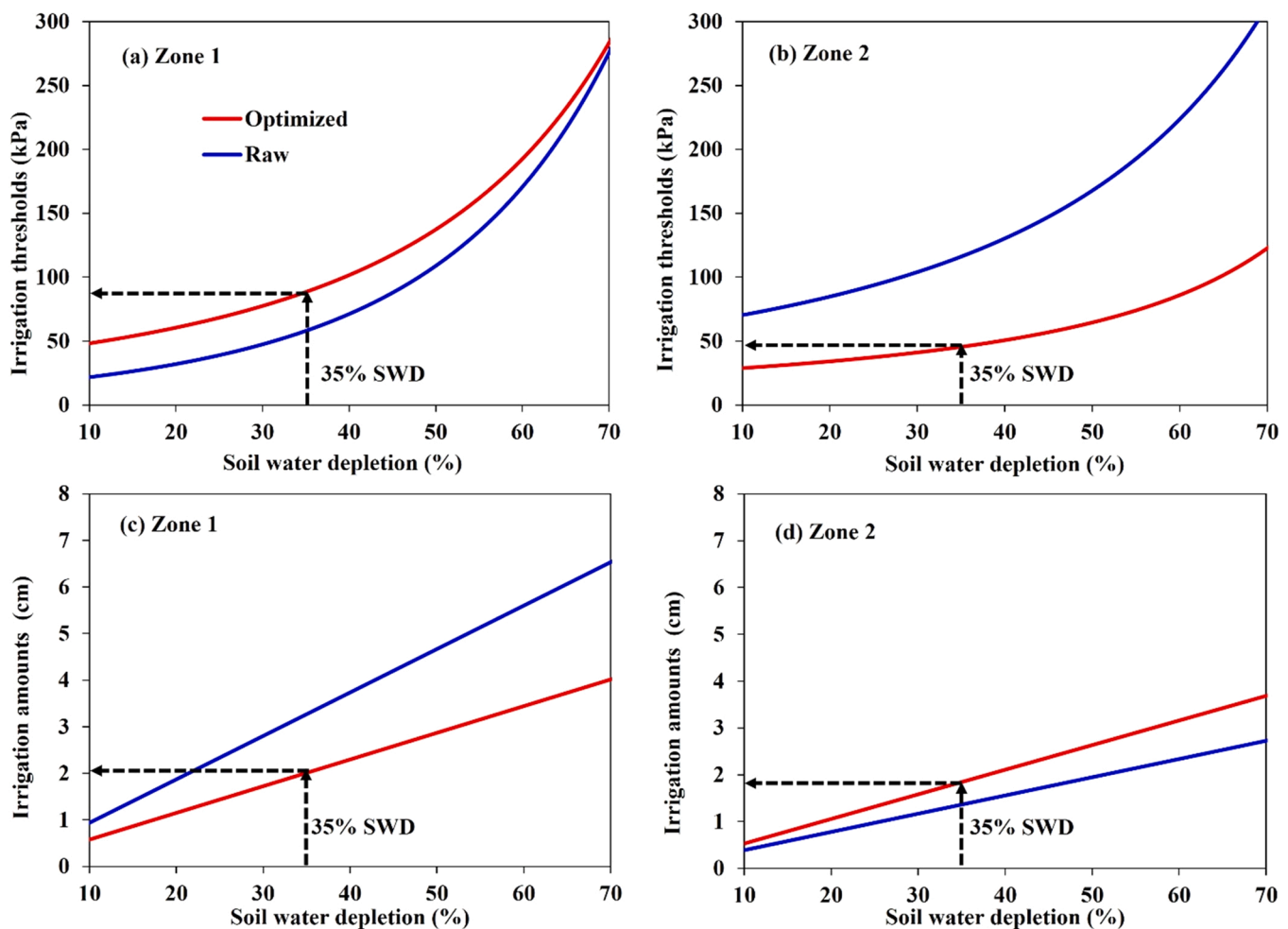


Fig. 5. Determined irrigation thresholds based on negligible drainage flux FC criterion in (a) zone 1 and (b) zone 2 using optimized and raw SHPs with optimized and raw FC. Irrigation amounts in (c) zone 1 and (d) zone 2 with corresponding soil water depletion variation using optimized and raw SHPs with optimized and raw FC. The black dotted arrows show the values of irrigation thresholds and amounts corresponding to 35% soil water depletion.

irrigation scheduling using raw SHPs and FCs. Therefore, it is essential to optimize zone-specific SHPs and FC to adopt best irrigation practices in each zone.

For example, using a hypothetical 35% SWD as the timing of irrigation value, the optimized soil matric potential (h_{35}) at 35% SWD was -88 kPa with a corresponding irrigation amount of 20 mm for a 60 cm soil profile in zone 1. On the other hand, the optimized h_{35} in zone 2 was -45 kPa, and its corresponding irrigation amount was 18 mm (Table 4). At SWD of 35%, the raw h_{35} using flux-based FCs in zone 1 and zone 2 was -58 kPa and -116 kPa, respectively. These differences in the optimized and raw irrigation thresholds at 35% SWD in both zones were due to discrepancies in SHPs and the site-specific measurements were incorporated in the case of optimized computations but not in the raw computations. From these findings, it was observed that the accurate determination of irrigation threshold and amount is highly dependent on accurate and effective SHPs, which showed zone-specific soil water dynamics and discrepancies in determination of FC values. As a result, irrigation scheduling using optimized parameters can be an efficient irrigation practice (Figs. 4a and 4b) as compared to raw parameters.

3.5. Determination of irrigation thresholds and amounts based on benchmark FC

Both zones had a similar pattern of irrigation thresholds and amounts variation with increasing SWD (Fig. 6). The irrigation thresholds as absolute values (Figs. 6a and 6b) and irrigation amounts (Figs. 6c and

6d) were lower considering benchmark FC and optimized SHPs in both zones. Using a hypothetical SWD of 35%, the optimized h_{35} using benchmark FC was -76 kPa and raw h_{35} using benchmark FC was -102 kPa (Table 4). However, in zone 2, optimized h_{35} using benchmark FC was -59 kPa and raw h_{35} using benchmark FC was -70 kPa (Table 4). The change in FC values resulted in different values of irrigation thresholds and amounts at the SWD. Based on the linear regression between accumulated ET_c and required irrigation using benchmark FC with optimized SHPs, the R^2 (0.5 in zone 1 and 0.4 in zone 2) was quite identical in both zones; however, it was lower than the flux-based criterion with optimized SHPs (Figs. 4a and 4b). Different SHPs can lead to various FCs and available water content values in the soils. Therefore, a correct zone-specific FC value can help in efficient zone-specific irrigation water management within the crop field. It can be concluded that the determination of FC in field conditions using optimization of SHPs can be an essential and foremost step in adopting best irrigation practices.

3.6. Limitations of the study and future research

This study presented an inverse modeling approach to optimize the zone-specific irrigation recommendations using zone-specific optimized flux-based FC. The irrigation thresholds and amounts using flux-based FC with optimized SHPs can potentially reduce the water stress in the crops, increase irrigation water use efficiency, and conserve water resources for irrigation. However, the limitation of this study is that we

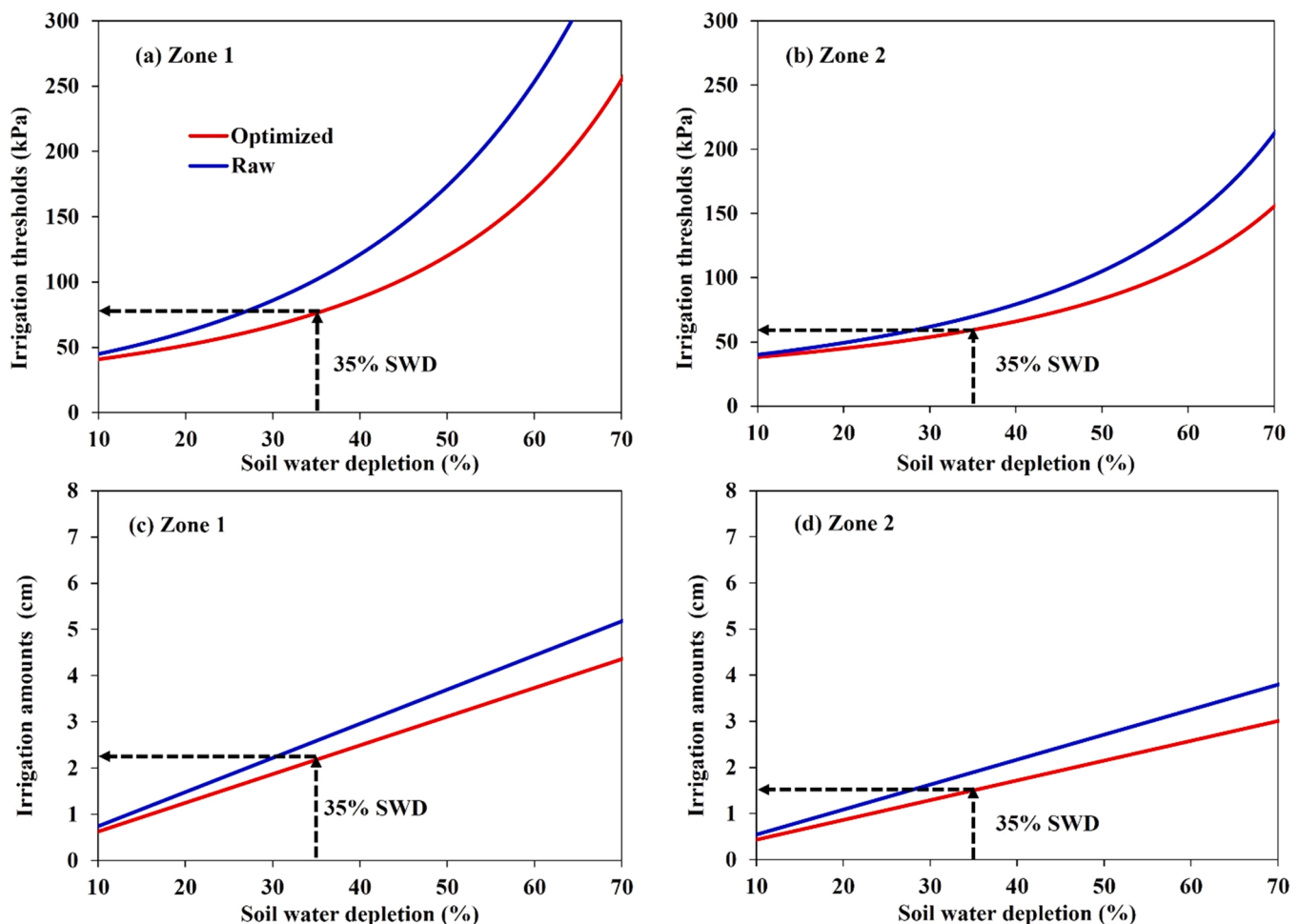


Fig. 6. Determined irrigation thresholds based on benchmark FC in (a) zone 1 and (b) zone 2 using optimized and raw SHPs with benchmark FC. Irrigation amounts in (c) zone 1 and (d) zone 2 with corresponding soil water depletion variation using optimized and raw SHPs with benchmark FC. The black dotted arrows show the values of irrigation thresholds and amounts corresponding to 35% soil water depletion.

need field measurements for inverse modeling to optimize FC and to determine the effective irrigation thresholds and amounts. This study is one-dimension using HYDRUS-1D, which does not account for lateral water flow in the model; hence, it would be interesting to incorporate two or three-dimensional models in the future. Irrigation recommendations using SHPs derived from conventional methods, such as Tempe cell/Sand kaolin box and compare with SHPs from HYPROP would be a future research question. Also, it would be interesting to measure site-specific ETc using lysimeters and generate irrigation recommendations. The outcomes of the study can be used in the crop models to understand the impact of irrigation recommendations on crop yield. This study used a simplified single-layer version of a complex subsurface system, and it would be of interest to investigate a multilayer complex model.

4. Conclusions

The flux-based optimized FCs yielded best result for the accumulated ETc and irrigation recommendations in both zones compared to irrigation recommendation using raw and benchmark FCs. The determination of irrigation thresholds was highly dependent on the determination of FCs and SHPs in each zone. The flux-based FC determination showed the importance of using optimized SHPs over the raw SHPs since raw and benchmark FCs in both zones did not consider soil water dynamics and zone-specific characteristics. The optimized FC was at -39 kPa in zone 1 and -25 kPa in zone 2. However, raw FCs in zone 1 and zone 2 were at -15 kPa and -59 kPa, respectively. The zone-specific optimized FC in absolute values show that raw and benchmark FC values were underrepresented in zone 1 and overrepresented in zone 2. Therefore, raw and benchmark FCs may not be the best options in irrigation scheduling schemes without optimization. Also, assuming a single FC (benchmark) throughout the field may not be the best option for irrigation recommendations. The determination of FC using zone-specific optimized and raw SHPs with a negligible drainage flux methods showed differences in irrigation thresholds and amounts to schedule irrigation resulting from the differences in the $K-h-\theta$ relation of SHPs. Hence, considering zone-specific optimized irrigation thresholds with optimized FC and SHPs and adopting a variable rate irrigation scheme for efficient irrigation water management in the zones can be the best climate-resilient irrigation solution.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

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